SEPARATION AND RECOVERY

I can predict what is produced. I cannot yet predict what becomes usable.

The gap between those two numbers may matter more than another small accuracy improvement.

PERSONAL NOTE FOR

Professor Gregory W. Severin

Michigan State University and FRIB Isotope Harvesting Group

I am a 12th grader at Glen Burnie High School, and I built a reduced model of the Ra-226 to Ac-225 production pathway. My equations follow irradiation, isotope buildup, decay, and cooling. They stop at the exact point where isotope harvesting and chemistry begin, which is why I am reaching out to your group.

THE RESULT THAT CHANGED HOW I SEE THE PROJECT

When I reconstructed the published Berkeley experiments, I had to keep produced activity and recovered activity as different quantities. The recovery was not identical between the two experiments. That made something obvious: a schedule that produces the most atoms is not automatically the schedule that delivers the most usable Ac-225, or does it fastest.

Produced in target	->	Separate and purify	->	Quality checked
What my model predicts		Time and losses unknown		The usable result I need

The practical numbers I am missing

- Time from end of irradiation to completed separation.
- Recovery fraction and how it should be defined.
- Losses during purification, transfers, and quality checks.
- The point when material is considered usable rather than merely produced.

What I can do without retraining V3

- Add a separate chemistry and recovery layer after the frozen physics model.
- Run sensitivity ranges instead of pretending one facility value is universal.
- Rank usable activity and total turnaround, then verify the nuclear part with the trusted solver.
- Clearly label every value as published, estimated, or still unknown.

What I am asking: Could you recommend one public isotope-harvesting or radiochemistry process example that reports separation time, recovery, purification losses, and quality-control steps? If a person in your group works more directly with those measurements, I would be grateful for the right name.